Notes: Drechsler, Savov, Schnabl, and Wang (JF, 2026)

Deposit Franchise Runs

Contents

1 Big picture

- **Question.** Why did a few regional banks fail in 2023 even though many banks had large mark-to-market losses, and how should we think about the interaction between interest rates, uninsured deposits, and financial stability?
- **Core idea.** The deposit franchise is an asset: banks pay deposit rates below market and earn a spread. But it is a *runnable* asset, because if depositors leave, the future spread disappears.
- **Key mechanism.** When rates rise, long-duration bank assets lose value, but the deposit franchise gains value because low-beta deposits become more profitable. If uninsured depositors run, the bank loses exactly the asset that was offsetting the asset-side losses.
- **What is new.** The paper shows that a run can occur even when bank assets are fully liquid. The runnable object is not illiquid loans; it is the uninsured portion of the deposit franchise.
- **Main theoretical result.** Deposit-franchise runs are more likely when interest rates are high, when the uninsured deposit share is high, and when uninsured deposit betas are low. The central risk variable is not uninsured deposits per se, but *low-beta uninsured deposits*.
- **Risk-management dilemma.** A bank can shorten asset duration to avoid runs when rates rise, but then it risks insolvency when rates fall because the deposit franchise shrinks and short-duration assets do not appreciate enough. Escaping this dilemma requires more capital.
- **Simple capital formula.** To avoid both high-rate run risk and low-rate insolvency for any rate realization, the bank needs capital covering the potential loss of the uninsured deposit franchise, in the amount $u(1 - \beta^U)$ per dollar of deposits.
- **Main empirical strategy.** The authors estimate bank-level asset losses, insured and uninsured deposit betas, deposit operating costs, and deposit franchise values from public call reports, then compare bank values *with* and *without* a run.
- **Main quantitative results.** Asset values fell by about 8% from December 2021 to February 2023, but the average bank's deposit franchise rose sharply and offset most of that loss absent a run. Fewer than 1% of banks appear exposed to deposit-franchise runs; the banks that stand out are SVB, Signature, and First Republic.
- **Policy result.** Holding high-quality liquid assets alone does not solve this problem. The paper argues for either reducing low-beta uninsured deposits or requiring capital against the uninsured deposit franchise, so that monetary policy is not constrained by run risk.
- **Economic takeaway.** The 2023 episode was not just about unrealized bond losses. It was about banks whose valuation had shifted from contractually protected assets to an uninsured franchise that could vanish if depositors left.

2 Data and measurement (key objects)

2.1 Sample, dates, and unit of observation

- **Main data.** U.S. call reports from June 2015 to March 2024.

- **Main sample.** Commercial banks with at least $1 billion in assets as of December 2021 and a significant domestic deposit base. The paper excludes broker-dealers, credit card banks, custodians, foreign-owned banks, and banks with deposits-to-assets below 65%.
- **Sample size.** 714 banks in the main sample, including 17 large banks.
- **Benchmark dates.**
  - December 2021: pre-hike baseline.
  - February 2023: just before the regional bank crisis.
  - February 2024: one year later.
- **Average balance-sheet facts.** The average bank funds itself mainly with domestic deposits (86% of liabilities), of which about 38% are uninsured, and has an average book equity ratio of about 10%.

2.2 What is observed and what is not

- **Observed directly.** Asset holdings by repricing-maturity bucket, deposit quantities, interest expense on deposits, noninterest expense, and uninsured deposit shares.
- **Not observed directly.** The market value of nontraded assets, deposit franchise values, insured versus uninsured deposit betas, insured versus uninsured deposit operating costs, and the bank's exposure to a self-fulfilling run equilibrium.
- **Why the model is needed.** Public data reveal the ingredients of run exposure but not the key latent object: the value of the insured and uninsured deposit franchise as interest rates move.

2.3 Asset losses

- **Method.** The paper matches call-report repricing-maturity bins to Bloomberg total-return indices. Treasury indices are used for non-real-estate assets; mortgage-backed-security indices are used for real-estate assets.
- **Identification assumption.** Book and market values are assumed equal in December 2021, before the hiking cycle. The paper checks this using reported fair values of securities.
- **Main result.** Average mark-to-market asset losses are:
  - 8.22% of December 2021 assets by February 2023,
  - 7.37% by February 2024.
- **Large banks.** Average losses are somewhat smaller:
  - 6.75% by February 2023,
  - 5.98% by February 2024.
- **Interpretation.** These losses are large relative to the roughly 10% book equity ratio, which is why ignoring the deposit franchise makes the system look close to insolvent.

2.4 Deposit betas

- **Two-step strategy.**
  - Step 1: estimate each bank's *overall* deposit beta from the change in its deposit rate (interest expense over deposits) divided by the change in the Fed funds rate.
  - Step 2: use the cross-sectional relationship between the overall beta and the uninsured deposit share to split the overall beta into insured and uninsured components.
- **Why the second step works.** Banks with more uninsured deposits systematically have higher observed betas, so the uninsured share contains information about how much of the overall beta comes from uninsured funding.
- **Average estimated betas.**
  - December 2021: insured 0.216, uninsured 0.332.
  - February 2023: insured 0.108, uninsured 0.372.
  - February 2024: insured 0.332, uninsured 0.571.
- **Interpretation.** Uninsured deposits pay more market-sensitive rates than insured deposits, but for the vulnerable banks the uninsured beta is still low enough that uninsured deposits are extremely profitable when rates rise.

2.5 Deposit operating costs

- **Method.** Following Hanson et al. (2015), the paper runs a hedonic regression of banks' net noninterest expense on different deposit categories while controlling for other balance-sheet items.
- **Why this matters.** Low deposit rates are not pure rents. Banks incur operating costs to service the franchise, and these costs must be netted out when valuing it.
- **Estimated average costs.**
  - Insured deposits: 1.494%,
  - Uninsured deposits: 0.954%,
  - Overall deposits: 1.303%.
- **Size pattern.** Insured deposit costs are lower at large banks; the overall deposit cost in the largest size quartile is about 1.145%.
- **Interpretation.** Insured deposits are cheaper in rate terms but more expensive to service. Uninsured deposits often have lower operating cost and therefore can generate especially valuable franchise income when their betas are low.

2.6 Deposit decay and discounting

- **Decay parameter.** The baseline uses $\delta = 10\%$ annual decay, meaning an average deposit life of about ten years.

- **Why this is conservative.** The paper values the franchise of *existing* deposits and assumes that future customers net of acquisition costs contribute zero expected franchise value on average.
- **Discount rate in the empirics.** Because a 10-year decay implies long cash-flow duration, the empirical implementation discounts franchise cash flows using the 10-year Treasury yield.

2.7 Three bank-value objects used in the empirics

- **No-franchise value:** $A(\tilde{r}) - D$.
- **Run value:** $A(\tilde{r}) - D + DF^I(\tilde{r})$, which assumes all uninsured deposits leave.
- **No-run value:** $A(\tilde{r}) - D + DF^I(\tilde{r}) + DF^U(1, \tilde{r})$, which assumes uninsured depositors stay.

Interpretation. The gap between run and no-run value is exactly the value of the uninsured deposit franchise. Banks are vulnerable when that gap is large relative to capital.

3 Models and empirical specifications (all equations + interpretation)

3.1 Timing, deposit base, pricing, and operating costs

Time is discrete. The bank starts with deposits D, split into insured and uninsured parts:

$$D^I = (1-u)D, \tag{1}$$
$$D^U = uD, \tag{2}$$

where $u \in [0,1]$ is the uninsured deposit share.

After the interest-rate shock, the bank sets deposit rates according to

$$r_d^I(\tilde{r}) = \beta^I \tilde{r}, \qquad r_d^U(\tilde{r}) = \beta^U \tilde{r}, \tag{3}$$

with overall beta

$$\beta = (1-u)\beta^I + u\beta^U. \tag{4}$$

Operating costs per dollar of deposits are c^I and c^U, with overall cost

$$c = (1-u)c^I + uc^U. \tag{5}$$

In periods $t \geq 1$, deposits decay exogenously at rate δ:

$$D_t = (1-\delta)D_{t-1}. \tag{6}$$

Interpretation.

- Deposit betas summarize banks' market power over deposit pricing. Low beta means the bank does *not* pass through market rates one-for-one to depositors.
- The deposit spread is $(1-\beta)\tilde{r}$. That spread rises when interest rates rise, which is why the franchise becomes more valuable in high-rate environments.
- The deposit franchise is therefore like a swap in which the bank pays fixed operating costs and receives a floating deposit spread.

3.2 Asset portfolio and asset duration

The bank holds a mix of short-duration assets and long-term fixed-rate assets. If l is the share of long-term assets and A is the initial market value of assets, then after the interest-rate shock:

$$A(\tilde{r}) = \left[l \left(\frac{r+\delta}{\tilde{r}+\delta} \right) + (1-l) \right] A. \tag{7}$$

The dollar duration of assets at the initial rate r is

$$T_A \equiv - \left. \frac{\partial A(\tilde{r})}{\partial \tilde{r}} \right|_{\tilde{r}=r} = \frac{lA}{r+\delta}. \tag{8}$$

Interpretation.

- Long-duration assets hedge the deposit franchise because asset values rise when rates fall, exactly when the franchise becomes less valuable.
- But that same hedge becomes fragile when the franchise is uninsured and can run.

3.3 Withdrawals and the run technology

At $t = 0$, after the shock, uninsured depositors may withdraw. Let λ denote the fraction of uninsured deposits that remain:

$$\widetilde{D}^U = \lambda D^U = u\lambda D. \tag{9}$$

The remaining fraction is an increasing function of the bank's solvency ratio v:

$$\lambda = \Lambda(v). \tag{10}$$

For tractability, the paper uses a step function around a run threshold $v^\star$:

$$\Lambda(v) = \begin{cases} 0, & v < v^\star, \\ 1, & v \geq v^\star. \end{cases} \tag{11}$$

Interpretation.

- If the bank looks sufficiently weak, all uninsured depositors run.
- This is a reduced-form way to capture a very concentrated, highly attentive uninsured depositor base, like the one at SVB.
- Insured depositors are assumed not to run in the baseline.

3.4 Valuing the deposit franchise

The market value of the bank is

$$V(\tilde{r}) = A(\tilde{r}) - L(\tilde{r}), \tag{1}$$

where $L(\tilde{r})$ is the market value of liabilities. The paper decomposes value as

$$V(\tilde{r}) = A(\tilde{r}) - D + DF(\tilde{r}), \tag{15}$$

where

$$DF(\widetilde{r}) = D - L(\widetilde{r}) \tag{16}$$

is the value of the deposit franchise.

Absent runs, the deposit franchise is

$$DF(\widetilde{r}) = \left(\frac{(1-\beta)\widetilde{r} - c}{\widetilde{r} + \delta}\right) D. \tag{17}$$

Its dollar duration is

$$T_{DF} \equiv -\left.\frac{\partial DF(\widetilde{r})}{\partial \widetilde{r}}\right|_{\widetilde{r}=r} = -\frac{c + (1-\beta)\delta}{(r+\delta)^2} D \leq 0. \tag{18}$$

With a non-flat term structure, the same formula uses the relevant long-term yield y:

$$DF = D\frac{(1-\beta)y - c}{y + \delta}. \tag{20}$$

Interpretation.

- The franchise is valuable when deposit spread exceeds operating cost: $(1-\beta)\widetilde{r} > c$.
- It has *negative duration*. When rates rise, the present value of future deposit spreads increases and the present value of operating costs falls.
- This is why the franchise can offset asset-side losses in a hiking cycle.

3.5 Deposit-franchise runs

The total franchise splits into insured and uninsured components:

$$DF(\lambda, \widetilde{r}) = DF^I(\widetilde{r}) + DF^U(\lambda, \widetilde{r}). \tag{22}$$

Because a run destroys the uninsured part of the franchise,

$$DF^U(\lambda, \widetilde{r}) = \lambda DF^U(1, \widetilde{r}). \tag{23}$$

The bank's solvency ratio at outflow level λ is

$$v(\lambda, \widetilde{r}) = v(0, \widetilde{r}) + \lambda u \left(\frac{(1-\beta^U)\widetilde{r} - c^U}{\widetilde{r} + \delta}\right), \tag{24}$$

where

$$v(0, \widetilde{r}) = \frac{A(\widetilde{r}) - D + DF^I(\widetilde{r})}{D} \tag{25}$$

is the solvency ratio when all uninsured depositors withdraw.

An equilibrium solves the fixed-point condition

$$\lambda = \Lambda(v(\lambda, \widetilde{r})). \tag{26}$$

Interpretation.

- A run is self-fulfilling because if uninsured depositors leave, the bank loses the uninsured franchise, which lowers bank value, which validates the decision to leave.
- The run does *not* require liquidation losses on loans. Assets are fully liquid in the baseline model.
- What matters is the value of the uninsured franchise relative to capital.

Proposition 2 in words.

- If the bank remains above the run threshold even after losing all uninsured deposits, there is a unique no-run equilibrium.
- If the bank falls below the threshold even when uninsured deposits stay, there is a unique run equilibrium.
- If the bank is solvent when uninsured deposits stay but insolvent when they leave, both run and no-run equilibria exist.

3.6 Interest rates, duration choice, and capital

The asset duration that perfectly hedges the *no-run* value of the bank against falling rates corresponds to the long-term asset share

$$l^{\text{Insolv}} = \frac{(1-\beta)\delta + c}{(1+e)(r+\delta)}, \tag{27}$$

where $e = (A-D)/D$ is equity per dollar of deposits.

To avoid a run at high rates, the bank must not choose duration that is too long. The maximum long-term share consistent with no-run at high rates is

$$l^{\text{Run}} = 1 - \frac{u + \beta^I(1-u) + v^\star}{1+e}. \tag{28}$$

The bank escapes the hedging dilemma if and only if initial value is high enough:

$$v(r) \geq v^\star + u(1-\beta^U). \tag{30}$$

With an upper bound $\bar{r}$ on future rates, this becomes

$$v(r) \geq v^\star + u\frac{(1-\beta^U)\bar{r} - c^U}{\bar{r}+\delta}. \tag{31}$$

The minimum solvency ratio that prevents runs at realized rate $\widetilde{r}$ is

$$\underline{v}(\widetilde{r}) = v^\star + u\frac{(1-\beta^U)\widetilde{r} - c^U}{\widetilde{r}+\delta}. \tag{33}$$

Interpretation.

- Long-duration assets hedge low-rate insolvency.
- Short-duration assets hedge high-rate run risk.
- Banks with a lot of low-beta uninsured deposits need extra capital because asset choice alone cannot protect them against both sides of the rate distribution.

3.7 Mark-to-market accounting and empirical bank-value objects

The paper distinguishes the market value of the bank from book-value equity marked only on the asset side:

$$v(\widetilde{r}) = e(\widetilde{r}) + \frac{DF(\widetilde{r})}{D}. \tag{29}$$

Interpretation.

- If one recognizes unrealized asset losses but ignores franchise gains, one understates solvency absent a run.
- But if one capitalizes the uninsured franchise fully, one understates liquidity risk because that franchise can disappear in a run.

Empirically, the paper evaluates three scenarios:

$$\begin{aligned} \text{No-franchise value} &= A(\widetilde{r}) - D, \\ \text{Run value} &= A(\widetilde{r}) - D + DF^I(\widetilde{r}), \\ \text{No-run value} &= A(\widetilde{r}) - D + DF^I(\widetilde{r}) + DF^U(1, \widetilde{r}). \end{aligned}$$

Interpretation.

- The no-franchise value asks how banks look if deposits are worth book value.
- The run value asks whether banks survive if all uninsured franchise value disappears.
- The no-run value asks whether the bank is solvent if depositors stay.

3.8 Extensions: rate-driven outflows, credit risk, and monetary policy

The baseline allows only run-driven outflows at $t = 0$. The paper adds a second type: rate-driven outflows to higher-yield outside options. If depositors withdraw a fraction $w(\widetilde{r})$ when rates move, then

$$D_0 = [1 - w(\widetilde{r})]D. \tag{34}$$

The franchise value becomes

$$DF(\widetilde{r}) = [1 - w(\widetilde{r})] \left(\frac{(1-\beta)\widetilde{r} - c}{\widetilde{r} + \delta} \right) D, \tag{35}$$

and the paper shows this is equivalent, for hedging purposes, to a higher effective deposit beta:

$$\widetilde{\beta} = \beta + w'(r)[(1-\beta)r - c] \left(1 + \frac{r}{\delta}\right). \tag{38}$$

Interpretation.

- Rate-driven outflows make the franchise less negative-duration because the bank keeps fewer deposits just when they become more profitable.
- In practical terms, the bank should hedge as if its deposit beta were higher.

The paper also extends the model to credit losses. Any negative credit shock that lowers $A(r, s)$ can push the bank into the multiple-equilibrium region, especially when rates are high and the uninsured franchise is large. This is why credit risk and interest-rate risk interact.

Finally, the paper studies monetary policy. If banks are well capitalized, the central bank can set rates without worrying about deposit-franchise runs. If banks are weakly capitalized, hikes can trigger runs when duration is long, while cuts can create insolvency when duration is short.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies asset losses

- **Source of variation.** Cross-bank differences in maturity structure and asset composition.
- **Why this is credible.** The mapping from maturity bins to market-value changes is transparent and close to Jiang et al. (2024), but the paper adds the deposit-franchise side rather than stopping at asset losses.
- **Validation.** The assumption that December 2021 book value equals market value is checked using reported fair values of securities.

4.2 What identifies deposit betas

- **Time-series variation.** Changes in deposit interest expense relative to changes in the Fed funds rate identify overall deposit pass-through.
- **Cross-sectional variation.** The uninsured deposit share predicts the overall beta. This is what identifies the gap between insured and uninsured betas.
- **Key assumption.** The insured-uninsured beta gap is constant across banks in the cross section at a given date.
- **Important empirical sign.** Figure 6 shows a strong positive relationship between deposit beta and the uninsured share in February 2023.

4.3 What identifies deposit operating costs

- **Approach.** A hedonic regression of net noninterest expense on deposit quantities and controls.
- **Why this is reasonable.** Conditional on controls, deposits should explain servicing costs; the paper also checks that the estimated relationship is not spuriously driven by concavity or selection.
- **External validation.** Bank of America's disclosed deposit costs line up closely with the paper's estimates.

4.4 What identifies run exposure

- **Core object.** The uninsured deposit franchise value relative to capital.

- **Why the framework matters.** A bank is exposed when the no-run value is positive but the run value is low enough that uninsured withdrawals can justify a run. Looking only at asset losses misses the buffer provided by insured deposit franchise value.
- **Cross-sectional discriminator.** The framework predicts vulnerability for banks with
  - high uninsured shares,
  - low uninsured betas,
  - long asset duration,
  - and limited capital.

4.5 Why insured versus uninsured deposits must be separated

- **Theoretical reason.** Insured and uninsured deposits have different run behavior and often different betas and operating costs.
- **Quantitative reason.** The paper finds that insured franchise value is large for most banks and is the main reason most banks remain solvent in the 2023 episode.
- **Why this changes the answer relative to some alternatives.** If one ignores insured franchise value and asks only whether market-valued assets cover withdrawals, many banks look fragile. If one includes insured franchise value, fragility becomes much more concentrated.

4.6 What makes the exercise credible

- **It fits the stock-market puzzle.** Banks had enormous unrealized asset losses, yet bank equity did not collapse during 2022. The model explains this because rising franchise value offset falling asset value.
- **It identifies the banks that failed.** SVB, Signature, and First Republic are clear outliers in the run/no-run framework.
- **It explains non-failures.** Other banks with large asset losses but either mostly insured deposits or high-beta uninsured deposits are not predicted to fail.
- **It is transparent about uncertainty.** The paper reports multiple dates, distinguishes run from no-run values, and does not rely on one knife-edge aggregate statistic.

4.7 Main limitations

- **Mapping uninsured to "flighty" and insured to "sleepy" is imperfect.** Some uninsured depositors may be inattentive; some insured depositors may still leave.
- **Public-data limitation.** The framework cannot directly measure bank-specific customer acquisition costs or internal runoff data; the decay parameter is calibrated from outside evidence.
- **Forward-looking beta issue.** Deposit betas in an ongoing hiking cycle may still be rising, so current measured betas can understate forward-looking franchise pass-through.

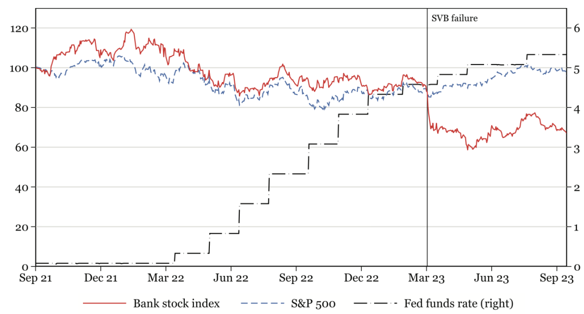

Figure 1. Bank equity index versus the market and the Fed funds rate. The figure plots the NASDAQ Bank Stock Index (BANK) in red, the S&P500 index in blue dash, and Fed funds rate in black dash-dot (right scale). The bank index and S&P 500 are indexed to 100 in September 2021. The vertical line marks the failure of Silicon Valley Bank. (Color figure can be viewed at wileyonlinelibrary.com)

- **Baseline no-fire-sale assumption.** The paper is intentionally isolating a new run mechanism, so it abstracts from liquidation costs. In practice, traditional run channels and deposit-franchise runs can coexist.
- **Partial-equilibrium focus in the bank cross section.** The exercise is not a full general-equilibrium crisis model with contagion, policy intervention, and endogenous macro conditions.

4.8 Relation to the literature

- **Bank runs.** Relative to Diamond-Dybvig type models, the paper's innovation is that the run can happen even with fully liquid assets because the franchise itself is runnable.
- **Interest-rate risk management.** The paper extends Drechsler, Savov, and Schnabl (2021), where long-duration assets hedge the negative duration of the deposit franchise, by allowing deposit outflows and showing how the hedge can fail.
- **2023 bank-crisis literature.** Relative to Jiang et al. (2024), the paper places much more weight on valuing both insured and uninsured franchise assets and therefore finds a much narrower set of vulnerable banks.
- **Customer capital.** The deposit franchise is treated as a type of customer capital, linking banking to the broader literature on indirect costs of financial distress.

5 Tables and figures

5.1 Figure 1: bank stocks, the market, and the Fed funds rate

Read:

- Figure 1 plots the NASDAQ bank index, the S&P 500, and the Fed funds rate from late 2021 through the 2023 crisis.
- The key fact is that bank stocks did *not* collapse during 2022 even as rates rose almost 5 percentage points.
- The break happens in March 2023, at the time of SVB's failure.

- This figure motivates the whole paper: if banks had really just absorbed an enormous unhedged duration shock, investors should have panicked much earlier.
- The model's answer is that rising deposit-franchise values were supporting bank values until depositors realized that the franchise itself could run.

5.2 Figure 2 and Figure 3: the run technology and the shift to multiple equilibria

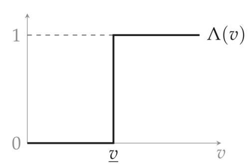

Figure 2. Fraction of remaining depositors $\Lambda(v)$ **as a function of the solvency ratio** v.

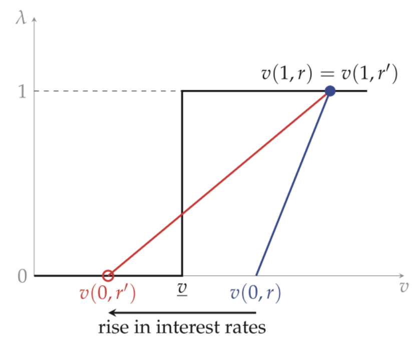

Figure 3. Unique equilibrium (blue dot) at rate r**, two equilibria (blue dot and red circle) at higher rate** $r' > r$. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- Figure 2 shows the step function $\Lambda(v)$: below the threshold, uninsured depositors run; above it, they stay.
- Figure 3 shows how an increase in interest rates can create multiplicity. The no-run value remains high because the uninsured franchise is valuable, but the run value falls because losing that franchise is now very costly.
- The visual lesson is that hikes increase the gap between the good equilibrium and the bad equilibrium.
- This is a new monetary-policy channel: tighter policy can create fragility not because assets become illiquid, but because the lost franchise becomes more important.

5.3 Figure 4 and Figure 5: the hedging dilemma and capital requirement

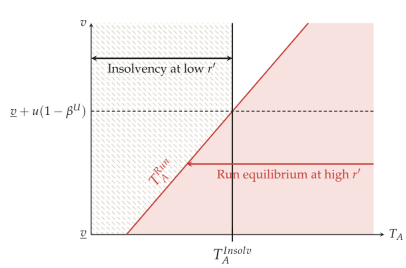

Figure 4. Risk management dilemma for weakly capitalized banks. The figure plots the regions in which (i) a run equilibrium exists at a sufficiently high interest rate (red) and (ii) the bank is insolvent at a sufficiently low interest rate (black) as a function of asset maturity T_A and bank value v. The dilemma occurs when $v < \underline{v} + u\left(1-\beta^U\right)$ (dashed line). In that case there is no asset maturity that avoids both run and insolvency risk. Banks with an uninsured deposit franchise ($u > 0$ and $\beta^U < 1$) thus need additional capital in the amount $u\left(1-\beta^U\right)$ to avoid instability. (Color figure can be viewed at wileyonlinelibrary.com)

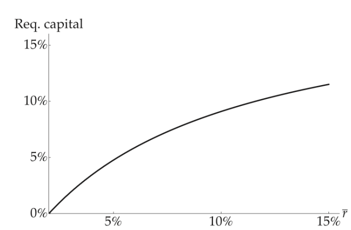

Figure 5. Capital that avoids a risk management dilemma as a function of the upper bound on rates $\bar{r}$.

Read:

- Figure 4 plots the two danger regions as a function of asset duration and bank value:
  - long duration creates run risk at high rates,
  - short duration creates insolvency risk at low rates.
- Weakly capitalized banks sit exactly where these two regions overlap, so there is no asset duration that protects them from both sides.
- Figure 5 translates the theory into a stress-test style capital schedule: the larger the upper bound on future rates, the more capital is needed to avoid the dilemma.
- These two figures are the cleanest policy takeaway in the theory section: the relevant capital requirement should scale with low-beta uninsured deposits.

5.4 Figure 6: why uninsured shares help identify uninsured betas

Read:

- Figure 6 is a binscatter of deposit beta against the uninsured deposit share in February 2023.
- Banks with more uninsured deposits have systematically higher overall deposit betas.
- This is the empirical fact that allows the authors to split overall betas into insured and uninsured components.
- The figure also supports the idea that uninsured balances are behaviorally different funding sources, not just accounting categories.

5.5 Table I: inputs to the empirical valuation

Read:

- Panel A reports mark-to-market asset losses: average bank loss of 8.22% by February 2023.
- Panel B reports insured and uninsured deposit betas:
  - insured betas are low, especially in early 2023,
  - uninsured betas are higher, but still well below one for many banks.
- Panel C reports deposit operating costs and shows that insured deposits are more costly to service.
- This table gives the full set of ingredients for franchise valuation. It is the empirical counterpart of the model's primitives.

5.6 Table II: bank values with and without the franchise

Read:

- **No-franchise value.** Average bank value falls from 10.24% of assets in December 2021 to just 2.02% in February 2023. Under this metric, 26.47% of banks have nonpositive value.

Table I

Inputs to Estimation

Panel A presents summary statistics for the mark-to-market asset losses for all banks and large banks. Asset losses are calculated using matched Bloomberg indices by asset class and repricing maturity. They are reported as a percentage of December 2021 asset values. Standard deviations are in parentheses. Panel B reports summary statistics for the estimated insured and uninsured betas, which are obtained under the assumption that they differ by a constant amount for each bank, obtained from the regression in Panel B of Table IA.II. Panel C summarizes the estimated insured, uninsured, and overall deposit costs for all banks and by size quartiles, calculated using estimates from Table IA.III.

Panel A: Mark-to-market asset losses

	All Banks			Large Banks		
	Dec 2021 (1)	Feb 2023 (2)	Feb 2024 (3)	Dec 2021 (4)	Feb 2023 (5)	Feb 2024 (6)
Asset loss	0.00	8.22	7.37	0.00	6.75	5.98
	(0.00)	(2.41)	(2.37)	(0.00)	(1.84)	(1.42)
Obs.	714	714	685	17	17	14

Panel B: Insured and uninsured deposit betas

	Dec 2021 (1)	Feb 2023 (2)	Feb 2024 (3)
Insured deposit beta	0.216	0.108	0.332
	(0.121)	(0.131)	(0.140)
Uninsured deposit beta	0.332	0.372	0.571
	(0.121)	(0.131)	(0.140)
Obs.	714	714	685

Panel C: Insured and uninsured deposit costs

	Insured (1)	Uninsured (2)	Overall (3)
All Banks	1.494	0.954	1.303
	(0.163)	(0.155)	(0.135)
Size			
1	1.602	0.946	1.376
	(0.026)	(0.135)	(0.082)
2	1.475	0.916	1.284
	(0.033)	(0.210)	(0.085)
3	1.656	0.945	1.406
	(0.056)	(0.141)	(0.119)
4	1.246	1.009	1.145
	(0.043)	(0.096)	(0.054)

Table II

Bank Values

The table reports our estimates of bank values. Panel A corresponds to all banks and Panel B to large banks. The first row is without the deposit franchise. Assets are marked to market using matched Bloomberg indices by asset class and repricing maturity. The second row is the bank value if a deposit franchise run occurs, $V(0,r) = A - D + DF_I$. The third value is if there is no run, $V(1,r) = A - D + DF_I + DF_U$. Deposit franchise values are calculated using the formula in Proposition 1 and our empirical estimates of deposit betas, costs, and the exogenous outflow rate (see Tables IA.II and IA.III). Bank values are reported on three dates, December 2021, February 2023, and February 2024. Bank values are scaled relative to assets. Standard deviations are in parentheses; % and # ≤ 0 are the percentage and number of banks with nonpositive value, respectively.

Panel A: All banks

	All Banks		
Bank Value	Dec 2021 (1)	Feb 2023 (2)	Feb 2024 (3)
$A - D$ (No DF)	10.24	2.02	2.86
	(2.07)	(3.20)	(3.18)
% ≤ 0	0.00	26.47	17.23
$V(0,r) = A - D + DF_I$ (Run)	8.81	9.76	8.02
	(2.33)	(3.76)	(3.53)
% ≤ 0	0.00	0.70	1.17
$V(1,r) = A - D + DF_I + DF_U$ (No run)	8.92	13.14	9.96
	(2.46)	(3.97)	(3.97)
% ≤ 0	0.00	0.14	0.73
Obs.	714	714	685

Panel B: Large banks

	Large Banks		
Bank Value	Dec 2021 (1)	Feb 2023 (2)	Feb 2024 (3)
$A - D$ (No DF)	9.96	2.95	4.20
	(1.68)	(2.57)	(1.83)
# ≤ 0	0	2	0
$V(0,r) = A - D + DF_I$ (Run)	9.80	8.50	9.17
	(1.62)	(4.27)	(2.34)
# ≤ 0	0	1	0
$V(1,r) = A - D + DF_I + DF_U$ (No run)	9.71	12.26	11.12
	(1.62)	(4.48)	(2.85)
# ≤ 0	0	0	0
Obs.	17	17	14

- **Run value.** Once insured franchise value is added, the average bank value is 9.76% in February 2023, and only 0.70% of banks have nonpositive value.
- **No-run value.** Adding the uninsured franchise raises the average bank value further to 13.14% in February 2023, with only 0.14% of banks nonpositive.
- **Large-bank panel.** Among the 17 large banks, the no-franchise metric flags two as negative in February 2023, but the no-run metric leaves all of them positive.
- The big lesson is that the deposit franchise completely changes the diagnosis. Ignoring it makes the system look broadly insolvent; including it makes fragility highly concentrated.

5.7 Figure 7 and Figure 8: who is actually vulnerable?

Read:

- Figure 7 plots bank values against the uninsured deposit share. In December 2021, the three value concepts are similar because the franchise is not yet valuable. By February 2023, they diverge sharply.
- Banks with low uninsured shares are cushioned by insured franchise value and are not run-vulnerable.
- Figure 8 zooms in on the 17 large banks. The striking outliers are SVB, Signature, and First Republic.
- These banks combine:

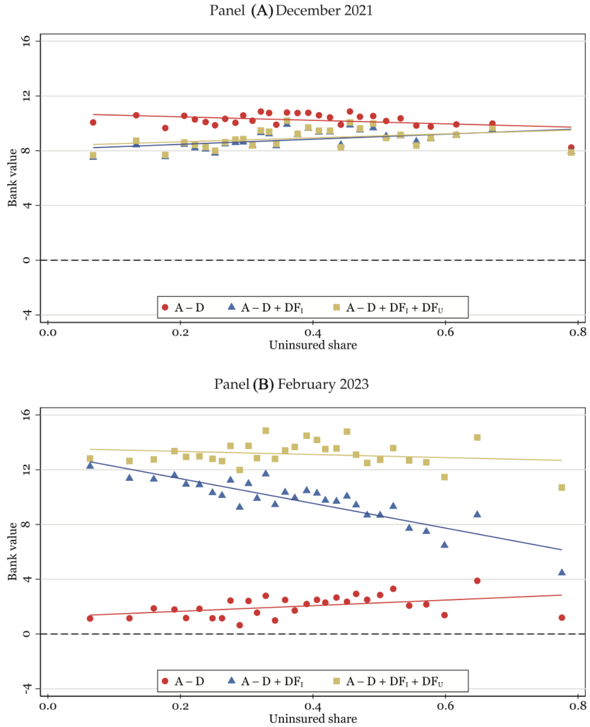

Figure 7. Bank values and the uninsured share of deposits. This figure plots bank values against the uninsured share of deposits as of December 2021 (Panel A) and February 2023 (Panel B). (February 2024 looks similar to February 2023.) Red circles show bank values without the deposit franchise, $A - D$. Blue circles show the value in a run, $V(0, r) = A - D + DF_I$. Yellow circles show the value without a run, $V(1, r) = A - D + DF_I + DF_U$. All values are scaled by assets. (Color figure can be viewed at wileyonlinelibrary.com)

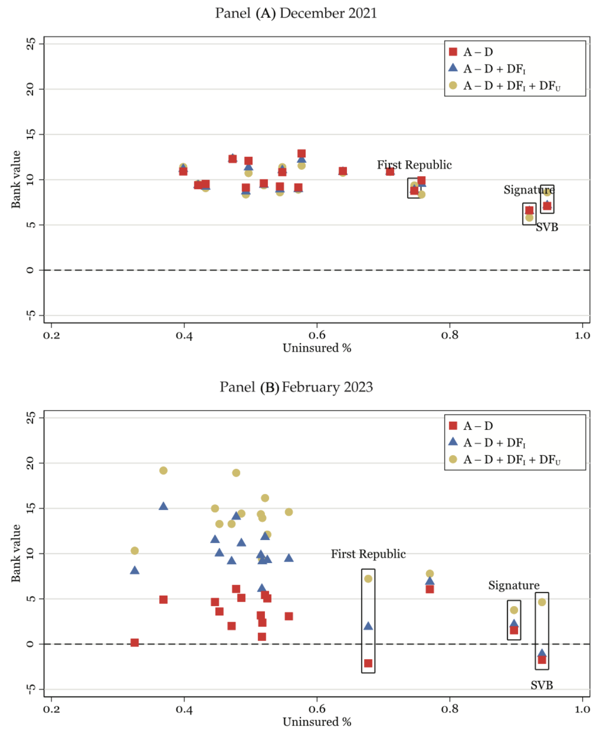

Figure 8. Large bank values. This figure plots bank values against the uninsured share of deposits, for the 17 large banks in our sample as of December 2021 (Panel A) and February 2023 (Panel B). Red circles show bank values without the deposit franchise, $A - D$, blue circles show the value in a run, $V(0, r) = A - D + DF_I$, and yellow circles show the value without a run, $V(1, r) = A - D + DF_I + DF_U$. All values are scaled by assets. (Color figure can be viewed at wiley-onlinelibrary.com)

  – very high uninsured shares,
  – low uninsured betas,
  – and relatively long asset duration.

- This is the figure that shows the model is not merely rationalizing the crisis after the fact. It sorts the banks that failed from those that did not.

5.8 Figure 9 and the extension section

Read:

- Figure 9 shows how a negative credit shock can shift the bank from a no-run equilibrium into

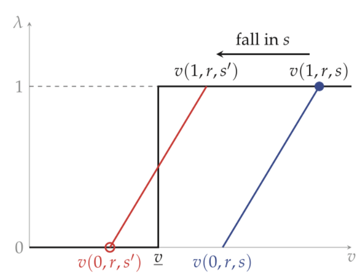

Figure 9. Effect of a credit loss $s' < s$. (Color figure can be viewed at wileyonlinelibrary.com)

the multiple-equilibrium region.

- The extension matters because it emphasizes that the deposit-franchise mechanism is not purely about interest rates. High rates make the bank vulnerable; other asset shocks can then trigger the run.
- The policy implication is that a bank can look hedged to rates on paper and still be fragile if the hedge relies on a runnable uninsured franchise.

6 Short summary

- **Method.** The paper combines a simple but sharp model of bank runs with a bank-level empirical valuation exercise based on call reports, deposit betas, operating costs, and repricing-maturity data.
- **Main conceptual result.** The deposit franchise is a runnable asset. This creates a run mechanism that does not require illiquid loans or fire sales.
- **Main quantitative result.** For the average bank, rising franchise value offset large asset losses in 2022–2023. Vulnerability was concentrated in a handful of banks with large low-beta uninsured franchises.
- **Policy lesson.** Capital and liquidity regulation should focus not just on uninsured deposits, and not just on asset liquidity, but specifically on the uninsured deposit franchise and the capital needed to survive its loss.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a new run channel: banks can suffer self-fulfilling runs because the uninsured portion of the deposit franchise disappears when depositors leave.
- It shows that this mechanism is especially powerful after interest-rate hikes, when the franchise has become valuable and asset values have fallen.
- It also shows empirically that this framework can explain why the 2023 regional bank crisis was concentrated in a few banks rather than broad based.

7.2 Economic intuition

- High interest rates do two opposite things to banks:
  - they reduce the value of long-duration assets,
  - and they increase the value of low-beta deposits.
- This looks like a hedge, and in normal times it is. But it is a hedge that depends on depositors staying.

- Once a bank's valuation relies heavily on an uninsured franchise, a run destroys the very asset that was offsetting the bond losses.
- That is why some banks could look fine to equity investors for most of 2022 and then fail very quickly in 2023.

7.3 Takeaway

This paper is important because it changes the way to think about bank fragility after large rate hikes. The right question is not just "How many unrealized losses do banks have?" It is "How much of the bank's value comes from an uninsured, low-beta deposit franchise that could vanish in a run?" In the authors' framework, that is the object that regulators should monitor, stress test, and capitalize. For understanding the 2023 regional bank crisis, that shift in perspective is the main contribution of the paper.